

Supporting Material: Artifact-Level Detail for a Condition-Indexed Operating Envelope and Representation-Repair Study in Automatic Modulation Classification

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Abstract

This is optional supporting material. It is written to be read on its own and is not cited by the manuscript it accompanies; the manuscript states every result it relies on. The purpose here is narrower: to expose the artifact-level detail behind those results, so that a reviewer who wants to audit the construction of the evaluation cells, the full capacity-control sweep, the per-SNR comparator-fidelity curves, or the seed-level robustness can do so directly. Nothing in this document introduces a new claim, a new experiment, or a new evidence class.

I. SCOPE AND HOW TO READ THIS DOCUMENT

The study behind this material asks when a compact spectral classifier and a much larger I/Q-domain classifier exchange rank under structured interference, describes that exchange with a two-variable empirical envelope, and then diagnoses and repairs the representation boundary the exchange reveals. All evidence is from source-disjoint 3GPP TR 38.901 TDL-profile simulation.

Because this document is standalone, the identifiers used throughout are restated here.

- **A0** is the direct spectral reference backbone (8,458 parameters) and **A5** is the sealed complete configuration (39,500 parameters), which adds a three-route spectral allocation, a simulation-component teacher, multi-task and contrastive objectives, a residual bypass, and a small route-orthogonality regularizer.
- **S**, **M**, and **L** are prospective spectral capacity tiers. *S* is the sealed A5 configuration; **M** and **L** widen only the spectral-channel, embedding, and environment dimensions, to 99,596 and 233,244 parameters.
- **sidecar** is a prospective received-I/Q repair branch that raises the total model size from 39,500 to 46,794 parameters.
- **IQFormer-inspired** and **MCLDNN** are architecture-faithful unified-retraining references, not exact reproductions of every original training recipe.
- The **target-support jammer overlap** o is the fraction of the jammer's total STFT power that falls inside the target's 99%-energy time-frequency support. It is not an occupied-bandwidth fraction.

Two evidence classes are kept apart everywhere. The *sealed* family is the original frozen confirmatory set; everything else, including the envelope, the severe-interference stratification, the Tier-2 interventions, and the capacity ladder, is *exploratory or post-hoc* and cannot change a sealed conclusion.

II. CONSTRUCTION OF THE EVALUATION CELLS

The envelope is fitted on aggregate cells rather than on individual windows. A cell is one (split, SIR stratum, overlap quartile) triple that retains at least 150 windows and all ten modulation classes.

TABLE I
CELL CONSTRUCTION FOR THE OPERATING ENVELOPE. “WINDOWS PER CELL” IS THE RANGE OF SOURCE-DISJOINT TEST WINDOWS CONTRIBUTING TO A CELL.

Split	Role	SIR strata (dB)	Cells	Windows per cell
id_test	fit	−10, −5, 0, +5, +10	16	230–1053
hard_interference	fit	−15, −10, −5, 0	16	290–343
unseen_jammer	held	−10, −5, 0, +5, +10	20	229–276
unseen_speed	held	−10, −5, 0, +5, +10	20	228–277
heldout_channel	held	−10, −5, 0, +5, +10	20	216–279
combined_ood	held	−10, −5, 0, +5, +10	20	223–270
clean_retention	excluded	0	1	5000
Fit total			32	
Primary held total			80	

TABLE II
HELD-REGIME ENVELOPE SKILL. RMSE VALUES IN PERCENTAGE POINTS.

Reference	Held regime	Cells	Env. Const.	R_{skill}^2	Oracle	Sign
IQFormer-insp.	unseen speed	20	3.94	7.87	0.75	6.71 1.000
IQFormer-insp.	held-out channel	20	4.86	7.59	0.59	6.95 1.000
IQFormer-insp.	combined OOD	20	7.50	9.23	0.34	4.24 0.950
IQFormer-insp.	unseen jammer	20	7.74	8.67	0.20	3.83 0.950
IQFormer-insp.	pooled (80)	80	6.23	8.37	0.45	6.04 0.975
MCLDNN	unseen speed	20	4.66	8.88	0.72	7.78 1.000
MCLDNN	held-out channel	20	5.41	9.06	0.64	8.41 1.000
MCLDNN	combined OOD	20	10.23	10.28	0.01	2.71 0.900
MCLDNN	unseen jammer	20	10.34	10.24	-0.02	3.23 0.900
MCLDNN	pooled (80)	80	8.10	9.63	0.29	6.81 0.950

Overlap quartile edges are computed within each split. Table I gives the resulting construction; the fit total of 32 and the primary held total of 80 are both reconstructible from its rows.

Two features of Table I are worth stating plainly. First, the `id_test` split contributes 16 cells rather than 20 because its lowest overlap quartile is populated almost entirely by jammer-free windows, which carry $o = 0$ and the $\text{SIR} = 0$ dB cache sentinel; that bin clears the 150-window floor only in the 0 dB stratum. Second, the single `clean_retention` cell is excluded from the primary envelope because a jammer-free window has no physical finite SIR, which is also why its window count is the full split rather than a quartile of it.

III. ENVELOPE SKILL BY HELD REGIME

Table II reports the held-regime skill for both strong references, including two diagnostics that are easy to omit and easy to misread. “Const.” is the fit-domain constant predictor: the mean gain over the 32 fitting cells, applied unchanged to the held cells, with $R_{\text{skill}}^2 = 1 - \text{MSE}_{\text{envelope}}/\text{MSE}_{\text{const}}$. “Oracle” is instead the mean of the held cells themselves, which no predictor could know in advance; it is reported because comparing against it shows what the envelope does *not* do.

The heterogeneity is real and is not smoothed away. Against IQFormer-inspired, the envelope transports the gain level well into the unseen-speed and held-out-channel regimes ($R_{\text{skill}}^2 = 0.75$ and 0.59) and much less well into the unseen-jammer and combined-OOD regimes (0.20 and 0.34); against MCLDNN the unseen-jammer and combined-OOD regimes are at or below the constant baseline. In every regime the oracle constant is competitive with, and often better than, the envelope, so the

TABLE III

CLEAN-SENTINEL FITTING-CELL SENSITIVITY. THE 31-CELL FIT EXCLUDES THE SINGLE $\text{id_test/SIR} = 0/\text{LOWEST-OVERLAP}$ AGGREGATE DOMINATED BY JAMMER-FREE SENTINEL WINDOWS; THE HELD EVALUATION IS THE UNCHANGED 80 FINITE-SIR INTERFERED CELLS. RMSE VALUES IN PERCENTAGE POINTS; BE DENOTES THE BREAK-EVEN OVERLAP.

Reference	Fit cells	Held RMSE	Const. RMSE	R^2_{skill}	Overlap coeff.	SIR coeff.	BE o @ -15	BE o @ -10
IQFormer-insp.	32	6.23	8.37	0.45	14.38	-1.036	0.11	0.47
IQFormer-insp.	31	6.42	8.28	0.40	15.90	-1.066	0.14	0.47
MCLDNN	32	8.10	9.63	0.29	19.16	-0.894	0.06	0.29
MCLDNN	31	8.20	9.59	0.27	20.55	-0.922	0.08	0.31

TABLE IV

ABSOLUTE MACRO-F1 ON THE MATCHED FIVE-SEED SUBSET.

Split	S	M	L	IQFormer-insp.	MCLDNN
hard interference	0.436	0.455	0.468	0.496	0.454
hard, SIR = -15 dB	0.353	0.384	0.403	0.262	0.255
in distribution	0.563	0.584	0.588	0.696	0.636
unseen jammer	0.456	0.462	0.466	0.612	0.584
unseen speed	0.487	0.507	0.510	0.626	0.582
held-out channel	0.488	0.510	0.515	0.613	0.567
combined OOD	0.450	0.462	0.465	0.621	0.587
clean retention	0.500	0.511	0.512	0.693	0.643

envelope’s demonstrated ability is transporting a gain *level* across regimes, not resolving variation within one.

IV. CLEAN-SENTINEL FITTING-CELL SENSITIVITY

The primary held envelope excludes jammer-free clean retention because a jammer-free window has no physical finite SIR. The fitting domain, however, retains one aggregate dominated by jammer-free sentinel windows, the $\text{id_test/SIR} = 0/\text{lowest-overlap}$ cell (mean overlap $\approx 10^{-11}$, 1053 windows, SIR taken from the cache sentinel). To close this definition asymmetry, we exclude exactly that aggregate, refit the two-variable envelope on the remaining 31 cells, and re-score the unchanged 80 finite-SIR interfered held cells; no model prediction changes and nothing is retrained.

The excluded aggregate is the $\text{id_test/SIR} = 0/\text{lowest-overlap}$ cell dominated by jammer-free sentinel windows. Excluding it moves the held RMSE by at most 0.20 percentage points and R^2_{skill} by at most 0.05, leaves both coefficient signs and the break-even ordering (required overlap rising as interference becomes less severe) unchanged, and therefore leaves the envelope conclusions materially unchanged; the 32-cell primary fit is retained.

V. CAPACITY CONTROL ACROSS ALL SPLITS

The capacity ladder isolates width from representation: S, M, and L differ only in three width numbers and share the routes, teacher, objective, and training recipe. All values in Tables IV and V are computed on the same matched five-seed subset, and must not be mixed numerically with ten-seed results reported elsewhere.

The pattern is consistent across splits. Width raises the average monotonically, and it helps most where the compact model was already competitive: at SIR = -15 dB the widest tier gains

TABLE V
PAIRED CAPACITY CONTRASTS ON THE MATCHED FIVE-SEED SUBSET, IN PERCENTAGE POINTS WITH 95% INTERVALS.

Contrast (hard interference unless noted)	Diff.	Low	High
M minus S	1.94	1.07	2.80
L minus S	3.18	2.44	3.96
L minus IQFormer-insp.	-2.84	-4.30	-1.31
L minus MCLDNN	1.37	0.03	2.72
L minus S, SIR = -15 dB	4.96	3.48	6.48
L minus IQFormer-insp., SIR = -15 dB	14.07	9.55	18.51
L minus IQFormer-insp., clean retention	-18.14	-19.34	-17.00

TABLE VI
PROSPECTIVE TIER-2 DIAGNOSIS, REPAIR, AND NEGATIVE CONTROL, RELATIVE TO A5, IN PERCENTAGE POINTS WITH 95% INTERVALS. EXPLORATORY THROUGHOUT.

Intervention	Params	Clean diff. [95% CI]	Hard diff. [95% CI]
Per-class clean coverage	39.5k	0.15 [-0.31, 0.63]	-0.30 [-0.91, 0.27]
Presence-gated route	39.5k	-0.30 [-0.66, 0.08]	-0.18 [-0.72, 0.35]
Received-I/Q sidecar	46.8k	11.77 [11.06, 12.49]	4.37 [3.44, 5.31]

4.96 percentage points over S and stays 14.07 percentage points ahead of IQFormer-inspired. It does not close the gap that matters for the boundary. On the marginal hard split L is still 2.84 percentage points behind IQFormer-inspired, and on clean retention it is 18.14 percentage points behind, a deficit essentially untouched by tripling the width from M to L (0.10 percentage points of movement). This is the arithmetic behind the manuscript’s statement that capacity is not a substitute for the representation change.

VI. PROSPECTIVE TIER-2 INTERVENTIONS

The Tier-2 chain was designed after inspecting the frozen evidence and sits outside the sealed family. Table VI restates its three arms with the intervals that the compact form in the manuscript omits.

The ordering carries the argument. Adding jammer-free windows for every class moves clean macro-F1 by 0.15 percentage points, so exposure alone is not the explanation. Gating the modulation route on predicted interference presence moves nothing either, and the learned gate saturates near the modulated route in both clean and hard conditions (99.994% and 99.991%), so it functions as a negative control rather than as a demonstration that a mask fallback was learned. Only restoring a small amount of received-I/Q information moves both conditions with intervals clear of zero.

VII. SEED-LEVEL ROBUSTNESS

The sealed intervals treat algorithm seeds as fixed blocks. Treating them instead as a random effect and resampling seeds directly gives [4.02, 5.19] percentage points for the A5–A0 whole-configuration contrast, [-0.15, 0.42] percentage points for the teacher-form contrast, [-0.15, 1.04] percentage points for the route-count contrast, and [-0.19, 0.62] percentage points for the residual-free A7 variant against A5. The qualitative reading is unchanged under either treatment: the whole-configuration effect is resolved, the isolated component effects are not.

TABLE VII
PER-SNR ACCURACY ON RML2016.10A OVER THREE SEEDS. BRACKETS ARE THE SEED MINIMUM AND MAXIMUM,
NOT CONFIDENCE INTERVALS.

SNR (dB)	MCLDNN mean	[min, max]	IQFormer-insp. mean	[min, max]
-20	12.1%	[12.0, 12.2]	14.9%	[14.5, 15.2]
-18	13.1%	[12.3, 13.9]	18.1%	[16.8, 19.0]
-16	14.8%	[14.0, 15.3]	22.3%	[21.7, 23.0]
-14	17.9%	[16.8, 18.9]	27.2%	[26.6, 28.3]
-12	22.0%	[20.4, 23.3]	33.3%	[31.5, 34.6]
-10	26.0%	[25.0, 26.7]	37.0%	[36.5, 37.8]
-8	32.5%	[31.1, 33.3]	41.4%	[41.1, 41.5]
-6	44.0%	[42.8, 45.7]	49.8%	[49.5, 50.3]
-4	58.8%	[58.3, 59.9]	63.3%	[63.0, 63.5]
-2	74.7%	[74.3, 75.0]	77.3%	[76.1, 77.9]
0	83.4%	[82.9, 84.1]	86.1%	[85.7, 86.7]
2	85.1%	[84.7, 85.6]	89.4%	[88.6, 90.5]
4	85.5%	[84.7, 87.1]	91.6%	[90.6, 93.0]
6	86.1%	[85.7, 86.8]	93.4%	[93.0, 94.1]
8	86.4%	[84.9, 88.6]	93.3%	[92.8, 94.2]
10	85.3%	[84.0, 87.1]	93.0%	[92.0, 94.4]
12	86.3%	[85.1, 88.4]	93.8%	[93.1, 94.7]
14	86.0%	[85.0, 87.9]	93.4%	[92.9, 93.9]
16	85.5%	[84.5, 87.1]	92.9%	[92.1, 93.7]
18	85.2%	[84.3, 86.8]	92.9%	[92.3, 93.9]
All SNR	58.55%	[58.1, 59.4]	65.23%	[64.8, 65.8]
Reference release	62.05%	—	64.19%	—

VIII. COMPARATOR FIDELITY ON A PUBLIC BENCHMARK

This check exists to establish that the two I/Q comparators are implemented competently, not to validate the simulator-domain envelope; a public benchmark cannot do the latter, because it does not expose the separately tracked target and jammer components the overlap covariate requires. The protocol follows the benchmark result table released with the IQFormer reference implementation: the 11-class RML2016.10a set, the full -20 to 18 dB grid, per-stratum 600/200/200 splits seeded at 233, three algorithm seeds, no augmentation and no input-length change.

Table VII makes the shape of the agreement explicit. Both implementations reproduce the reference release at the level of the all-SNR range, and IQFormer-inspired matches it closely (+1.04 percentage points), while MCLDNN remains 3.50 percentage points lower but within the same broad range. Agreement is not pointwise: individual SNR bins deviate by up to about 8 percentage points, which is why this is reported as range-level implementation fidelity rather than reproduction. No fidelity threshold was declared before these runs, so no pass/fail band is claimed here.

IX. ARTIFACT INDEX

Every number in this document is derived from the frozen artifact tree rather than transcribed from an earlier draft. The relevant entry points are:

- Cell construction and envelope fits: `analysis_zero_compute/outputs/csv/a14_envelope_cells.csv` and `a14_v41_envelope_interfered_holdout.csv`. The V4.4 clean-sentinel 31-cell sensitivity is recomputed from the same cell table by `analysis_zero_compute/a16_clean_sentinel_sensitivity.py`, writing `a16_clean_sentinel_sensitivity.csv` and `a16_clean_sentinel_sensitivity.json`.

- Capacity ladder: `tier2_capacity_L_v1_vs_S.csv`, `tier2_capacity_L_v1_vs_M.csv`, `tier2_capacity_L_v1_vs_iqformer.csv`, `tier2_capacity_L_v1_vs_mclnncnn.csv`, and `tier2_capacity_M_v2_vs_a5.csv`, all five-seed matched.
- Tier-2 interventions: `tier2_iq_sidecar_v1_vs_a5.csv`, `tier2_condition_coverage_v1_vs_a0.csv`, and `tier2_presence_gated_v1_vs_a5.csv`.
- Comparator fidelity: `artifacts/rml2016_10a_fidelity_v3/`, whose `run.json` records the dataset SHA-256, the split protocol, the reference-release values, and the environment.
- The overlap covariate is computed at cache-build time, stored as `overlap.npy` per split, and SHA-256 checksummed in the sealed cache manifest; it is never an input feature of any model.

The reproduction package released with the paper contains the resolved macros, the figure-source CSV files, per-seed aggregates, run configurations, and the full provenance manifest.